DO NOW — RECALL

Paper 2 extended response challenge

Name: _____ Class: _____ Date: _____

1. What is a precondition?

2. What is a loop invariant?

3. What does O(1) auxiliary space mean?

4. For [1,1,2,3,3], count globally distinct values and consecutive runs.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Requirement that must be satisfied for correctness.

2. Model answer

Property maintained as iterations progress.

3. Model answer

Additional memory remains constant with input size.

4. Model answer

Three distinct values; three consecutive runs.

By the end, you can…

• State the sorted-input precondition and any edge cases.

• Trace changing values with a loop invariant and correct bounds.

• Explain why the selected algorithm is efficient and when another is preferable.

Vocabulary — know these terms

precondition — Required assumption about input.

invariant — Property maintained by a loop.

auxiliary space — Memory used beyond input.

hash set — Structure for expected efficient membership tests.

NEW LEARNING

Read, annotate and explain

Step 1

State the sorted-input precondition and any edge cases.

Step 2

Trace changing values with a loop invariant and correct bounds.

Step 3

Explain why the selected algorithm is efficient and when another is preferable.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace UniqueCount for [1,1,2,3,3].

Explain the worked method

The loop starts at index1 and compares each item with its predecessor.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write pseudocode to count distinct values in a sorted array, including empty input. [8]

Question 2

Trace [2,2,5,5,6,9,9] and give time/space complexities. [9]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate adjacent counting versus a hash set when the data is unsorted and contains a million elements. [9]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An adjacent comparison algorithm for sorted input works identically on unsorted input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.